Biliary Atresia, Part Four: Timing of the Kasai Operation. Age Thresholds, Screening Programmes and Centralisation of Care.

Imagine a seven-week-old girl referred on a Thursday afternoon. At her two-week check she was a little yellow, but she was breastfed and thriving, so the family was told to come back if the jaundice did not settle. Her mother has now noticed that the stools are pale. The referring doctor asks a reasonable question: the ultrasound department can scan her next Wednesday, is that soon enough? Behind this question sits a larger one that every hepatologist and every paediatric surgeon has to answer. Does a week really matter in biliary atresia, and if it does, who controls the clock?

The main idea of this chapter can be said in three sentences. First, the chance of keeping the native liver falls steadily as the age at the Kasai operation rises; there is no age at which it is safe to delay, and no age at which the operation becomes pointless. Second, the age at surgery is set mostly before the infant ever meets a surgeon, by how quickly jaundice is recognised, how quickly the infant is referred, and how quickly the work-up is done. Third, two system-level tools change the outcome: screening, which moves the operation earlier, and centralisation of care in experienced centres, which improves what is done at any age.

A word on terms before we begin. The Kasai portoenterostomy is the operation in which the surgeon removes the scarred remnant of the bile ducts at the porta hepatis, the hilum of the liver where the ducts normally leave it, and sews a loop of small bowel, called a Roux loop, directly onto the cut surface, so that bile can drain from microscopic ducts into the intestine. Native-liver survival means being alive with one's own liver, without a transplant. Jaundice clearance means that the bilirubin falls to normal after the operation, usually defined as below twenty micromoles per litre, or about 1.2 milligrams per decilitre.

The sources for this chapter are national cohort studies from France, Canada, England and Taiwan; two meta-analyses, by Yang and colleagues in 2022 and by Hoshino and colleagues in 2023; a large American database study of age at surgery published in 2021; the Texas newborn screening study by Harpavat and colleagues in the Journal of the American Medical Association in 2020; a 2024 meta-analysis of screening strategies by Hari Gopal and colleagues; and the guidance documents: the 2017 joint guideline on cholestatic jaundice in infants from the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition and the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, known as NASPGHAN and ESPGHAN; the 2025 clinical report from the American Academy of Pediatrics, known as the AAP; and the 2026 consultation of the United Kingdom National Screening Committee. Neither the American Association for the Study of Liver Diseases nor the European Association for

the Study of the Liver has a guideline on the timing of the Kasai operation.

Part one. Why the clock runs.

To understand why timing matters, it helps to follow the disease step by step. The first step is that bile flow from the liver stops. In many infants this begins before birth. We know this because conjugated bilirubin, the water-soluble form of bilirubin that the liver has already processed, is often abnormal in the first days of life in infants who later prove to have atresia, long before the stools turn pale or the jaundice is obvious.

The second step is retention of bile. Bile acids are detergents; when they cannot leave the liver they injure the hepatocytes and the cells lining the small bile ducts, called cholangiocytes.

The third step is the liver's response. The liver grows new small ductules around the portal tracts, a pattern called the ductular reaction, and it activates hepatic stellate cells, which turn into scar-producing cells and lay down collagen around the portal tracts.

The fourth step is the one that matters most for the surgeon. The Kasai operation does not reconnect a bile duct. It opens the fibrous plate at the porta so that bile from tiny residual ductules can seep into the Roux loop. As the weeks pass, these ductules are destroyed and the ducts inside the liver become fibrotic, so there is less and less to drain.

The fifth step is cirrhosis. The scar bridges from one portal tract to another, the architecture of the liver is lost, and the pressure in the portal vein rises.

Put these steps together and the logic is plain: each week of obstruction removes drainable ductules and adds fibrosis, so both the chance of bile drainage and the long-term chance of keeping the liver fall. Because the process is continuous, there is no biological cliff at thirty, forty-five or sixty days. Those numbers are useful conventions for audit and for setting targets, not switches in the disease.

Part two. What the cohorts show.

The longest follow-up comes from the French national registry. In 2019 Fanna and colleagues reported 1,428 infants with biliary atresia born in France between 1986 and 2015, of whom 1,340 had a Kasai. Survival with the native liver was 41 percent at five years, 35 percent at ten years, 26 percent at twenty years and 22 percent at thirty years. When the authors divided the infants by the month of life in which they were operated, survival with the native liver twenty-five years later was 38 percent for those operated in the first month, 27 percent in the second month, 22 percent in the third month and 19 percent after the third month. In words, operating in the first month doubled the chance of reaching adulthood with one's own liver compared with operating after three months. Two other facts from that study are worth remembering. The median age at the Kasai was 59

days and did not change over thirty years, and overall survival improved over the same period, from 72 percent at five years in the oldest cohort to 87 percent in the newer ones, largely because of transplantation.

An earlier French report by Serinet and colleagues in 2009 followed 743 children. Survival with the native liver was 57.1 percent at two years, 37.9 percent at five, 32.4 percent at ten and 28.5 percent at fifteen years. Survival fell with each fifteen-day band of age at surgery, from thirty days or less up to ninety days, and this gradient held at every point of follow-up. The authors estimated that if every infant in France had been operated before forty-six days of age, about 5.7 percent of all paediatric liver transplants in the country each year could have been avoided.

A Canadian national study by Schreiber and colleagues in 2007, with 349 infants, showed the same shape in a different health system. At four years, 49 percent of infants operated at thirty days or less still had their own liver, compared with 36 percent of those operated between thirty-one and ninety days, and 23 percent of those operated after ninety days.

Two meta-analyses pool these cohorts. Hoshino and colleagues, in the European Journal of Pediatrics in 2023, combined nine studies. A Kasai at thirty days or less made survival with the native liver about one fifth more likely than a later operation, a risk ratio of 1.22, with a confidence interval from 1.13 to 1.31. Even compared only with infants operated between thirty-one and sixty days, the first month still helped, with a risk ratio of 1.13. Later surgery was associated with faster progression to transplantation, with a hazard ratio of 2.12, meaning that at any given time the rate of needing a transplant was roughly doubled. Yang and colleagues, in Frontiers in Surgery in 2022, combined fourteen cohort studies with 3,276 patients. Operating by ninety days roughly tripled the odds of jaundice clearance, an odds ratio of 3.05, and raised the odds of native-liver survival by about 70 percent. Operating by sixty days, compared with sixty-one to ninety days, raised the odds of native-liver survival by about 40 percent, an odds ratio of 1.41, but in this comparison the difference in jaundice clearance, an odds ratio of 1.31, was not statistically significant.

How should an attending read these numbers? The direction of the effect is consistent across every country and every analysis. The size of the effect for any single week is modest, and the effect is largest at the extremes. And all the data are observational; no one will ever randomise infants to a delayed operation, so some confounding will always remain.

Part three. Age is also a proxy.

The age at surgery measures more than time. Several features of the disease and of the health system travel with age, and they can make the age effect look larger or smaller than it really is.

The first is phenotype. In the English national series, infants with other major anomalies, mainly the syndromic form of atresia with splenic malformations, cleared their jaundice in 43 percent of cases compared with 57 percent of those with isolated atresia. Cystic atresia, in contrast, often presents early because the cyst is seen on prenatal or early scans, and it does relatively well. The second is anatomy: when the atresia leaves a patent common or hepatic duct, the forms called types one and two, drainage is better at any age than in the common type three, in which the obstruction reaches the porta. The third is the centre: an experienced surgeon and team shift the whole age curve upwards. The fourth is access: a late operation often marks late recognition, and in England infants of South Asian ancestry reached surgery later than others. The fifth is severity: two infants of fifty days can have very different degrees of fibrosis.

The practical conclusion is simple. Age is the one prognostic factor the system can change, and all the evidence points the same way. But age should be used to set urgency, not to predict an individual child's outcome.

Part four. The older infant.

What about the infant who is already three or four months old? The same studies answer this. In France, 19 percent of infants operated after three months still had their own liver twenty-five years later. In Canada, 23 percent of those operated after ninety days still had their own liver at four years. Later surgery shortens the time to transplant, but for many children it postpones the transplant rather than making it inevitable at once.

Most centres therefore still offer a Kasai to an older infant whose liver is not yet decompensated, meaning no ascites, synthetic function that corrects with vitamin K, and no established cirrhosis on imaging or biopsy. Even a partly working Kasai lets an infant grow to a size and age at which transplantation is safer and donor options are wider. A Chinese review in the World Journal of Pediatric Surgery in 2018 put the principle clearly: the Kasai should come first unless there is advanced liver failure at presentation. When the infant arrives with ascites, a coagulopathy that does not correct with vitamin K and a cirrhotic liver, a primary transplant assessment is the better path. The decision should rest on the state of the liver, not on the calendar.

Part five. Where the real-world clock stands.

How old are infants when they actually reach the operating room? A study of the Pediatric Health Information System database, published in the Journal of Pediatric Surgery in 2021, looked at 470 infants who had a Kasai at 46 American children's hospitals between 2016 and 2019. The median age was 57 days, with the middle half of infants operated between 42 and 72 days. Almost half, 212 of 470 or 45.1 percent, were operated at sixty days or later. The median age at individual hospitals ranged from 38 days at the fastest to

76 days at the slowest. A five-week spread between hospitals in the same country is a systems problem, not biology.

Europe is no faster. Across 14 European referral centres the median age was 61 days, against a stated goal of operating in the first four weeks of life. In England after centralisation the median was 54 days, and in France it has been about 59 days for thirty years. Without screening, the median infant everywhere reaches surgery at around eight weeks.

Part six. Screening.

There are two ways to screen a whole population of newborns for biliary atresia. The first is the stool colour card, a card with numbered photographs of normal and pale stools that parents compare with their infant's nappy, calling a hotline if the stool matches a pale picture. The second is measuring direct or conjugated bilirubin in blood in the first days of life, with a repeat test for those above the reference range.

Taiwan introduced the stool card nationally, printed in the child health booklet, and Lien and colleagues reported the results in Hepatology in 2011. They compared 89 infants born before screening, between 1990 and 2000, with 28 infants born during regional screening and 74 during universal screening, between 2002 and 2005. The proportion of infants who had their Kasai before sixty days rose from 49.4 to 65.7 percent. The proportion free of jaundice three months after surgery rose from 34.8 to 60.8 percent. Jaundice-free survival with the native liver at three years rose from 31.5 to 56.9 percent, and overall survival at five years rose from 55.7 to 89.3 percent. In the first screened cohort, jaundice-free native-liver survival at five years was 64.3 percent compared with 27.3 percent before screening. The design was a before-and-after comparison, and other aspects of care also improved over the decade, but the whole chain of benefit followed the shift in age.

Stool cards are cheap but imperfect, because they depend on parents and on a call-back system that works. In British Columbia, Woolfson and colleagues, in 2018, evaluated the first two years of a province-wide programme covering 87,583 births. The card reached 94 percent of families and cost about 68 cents per birth. But sensitivity was only 50 percent: of six infants with atresia, the programme as a whole succeeded for three. When it worked, the median age at Kasai was 49 days; when it failed, it was 116 days.

Blood screening is more sensitive because direct bilirubin is high from the first days, before the stools turn pale. In the Texas study by Harpavat and colleagues, 124,385 newborns at fourteen hospitals had a direct or conjugated bilirubin measured within the first sixty hours of life. Those above the laboratory's reference range were retested at or before the two-week visit, and the repeat was positive if it was higher than the first value or above one milligram per decilitre. All seven infants with atresia were found. Sensitivity

was 100 percent, although with only seven cases the confidence interval ran from 56 to 100 percent. Specificity was 99.9 percent and the positive predictive value was 5.9 percent, which means that about sixteen healthy infants were recalled for every true case. The mean age at the Kasai at Texas Children's Hospital fell from 56 days before screening to 36 days after, a difference of nineteen to twenty days.

The 2024 meta-analysis by Hari Gopal and colleagues in PLOS ONE pooled these strategies. Across five studies and 662,141 infants, direct or conjugated bilirubin screening had a pooled sensitivity of 100 percent and a specificity of 98.8 percent, with high certainty of evidence. Across seven studies and 996,262 infants, the stool colour card at one month had a pooled sensitivity of 79.6 percent and a specificity of 99.9 percent, with moderate certainty. In words, the card misses about one infant in five, and the blood test misses almost none.

Even so, policy is divided. In February 2026 the United Kingdom National Screening Committee opened a consultation on an evidence summary concluding that there is not enough evidence to recommend screening newborns for biliary atresia, judging test accuracy uncertain and noting that the age at surgery in the United Kingdom is similar to that in some countries that screen.

Part seven. Centralisation.

The second lever acts after referral. In 2000, McKiernan, Baker and Kelly reported in the Lancet every case of biliary atresia in the United Kingdom and Ireland over two years, from 1993 to 1995. Ninety-three cases were confirmed, a frequency of about one in 16,700 live births, and 91 infants were operated at fifteen different surgical centres. In centres treating more than five cases a year, 61.3 percent of children were alive with their native liver at five years. In centres treating fewer than five cases a year, only 13.7 percent were. Overall five-year survival was 91.2 percent in the larger centres and 75 percent in the smaller ones. Strikingly, once the size of the centre was accounted for, age at surgery no longer predicted survival in this series. In a fragmented system, the surgeon outweighed the calendar. The authors concluded that infants with atresia should be managed in centres treating more than five cases a year.

England and Wales acted on this. From 1999, every suspected case was referred to one of three centres, King's College Hospital in London, Birmingham Children's Hospital and Leeds, each able to perform both the Kasai and liver transplantation. Davenport and colleagues reported the first results in the Lancet in 2004: of 148 infants treated between 1999 and 2002, 57 percent cleared their jaundice, 51 percent had their native liver at four years, and 89 percent were alive. The national benchmark paper in 2011 described 443 infants born between 1999 and 2009. The median age at Kasai was 54 days, 55 percent cleared their jaundice, native-liver survival was 46 percent at five years and 40 percent at

ten, and true patient survival was 90 percent at five years and 89 percent at ten. For comparison, a 2025 European review reports native-liver survival at two years of only 29 percent in Germany before centralisation, after 160 Kasai operations between 2010 and 2014, and 70 percent at five years in Finland after its national centralisation programme. These comparisons are rough, because eras, time points and definitions differ, but the message is consistent.

It is worth being precise about what centralisation can and cannot do. It improves the technique of the portal dissection and the recognition of cystic and syndromic forms. It improves post-operative care, especially the prevention and treatment of cholangitis and nutritional support with fat-soluble vitamins. It lets the same team time the transplant without inter-hospital delay, and it creates a registry for audit. What it does not do is find the infant earlier: the English median age stayed in the mid-fifties. France offers a counterpoint: it has kept a decentralised network, and the effect of centre caseload was significant only in its earliest era, from 1986 to 1996, suggesting that networks and shared protocols can partly substitute for concentration. Screening shortens the time before the surgeon; centralisation improves what the surgeon and the team do. Neither replaces the other.

Part eight. A clock measured in days, and what the guidelines say.

Working backwards from a Kasai before thirty to forty-five days gives a concrete timetable. Where newborn screening exists, the direct bilirubin from the first days is the first data point, and any value above the reference range should be noted. At the two-week visit, the clinician looks for jaundice, looks at the stool personally, and reviews any previous direct bilirubin; if any is abnormal, a direct bilirubin is measured that day. A direct bilirubin of one milligram per decilitre or more leads to contact with paediatric hepatology in the same week, vitamin K, and a work-up done in parallel rather than one test after another: ultrasound, gamma-glutamyl transferase, matrix metalloproteinase seven, and alpha-1 antitrypsin level and phenotype, with a liver biopsy when the picture is unclear. Then the cholangiogram and the Kasai are done in a single anaesthetic in an experienced centre.

How do the guidelines line up? The 2017 NASPGHAN and ESPGHAN guideline asks for a fractionated bilirubin in any infant still jaundiced at two weeks of age, allowing breastfed infants who are otherwise well to be reviewed at three weeks. The 2025 AAP clinical report, led by Harpavat, sets a three-step strategy for primary care at two to four weeks: test the direct bilirubin if there is jaundice, if there are pale stools at any time, or if a previous direct bilirubin was high. It treats a direct bilirubin of one milligram per decilitre or more as a reason for urgent consultation, recommends scheduling the well-child visit at two weeks, and names a Kasai before thirty days as the aim. In the

United Kingdom, the National Institute for Health and Care Excellence uses a conjugated bilirubin above twenty-five micromoles per litre at the prolonged-jaundice check. On when to test and when to refer, these documents are aligned. On the target age they are also aligned, since European centres state the first four weeks as their goal, although the median they achieve is about sixty days.

They differ on screening. The AAP describes universal screening as a future goal; Taiwan, Japan and Switzerland run national stool card programmes; and the United Kingdom committee does not recommend screening. The most material difference is centralisation: England, the Nordic countries and the Netherlands concentrate care in a few centres, while North America has no such requirement. Finally, none of the North American documents addresses the older infant, which remains a gap filled by expert practice. The overall picture is that North America has set an ambitious age target without either the screening or the centralisation that brought earlier and better surgery elsewhere, and almost half of American infants still miss sixty days.

Part nine. Traps, and what to do differently in clinic.

Several reasoning errors reliably cost weeks. The first is to say that a breastfed baby can be rechecked in two weeks. Breast milk jaundice is unconjugated, and the only way to know is to fractionate the bilirubin at the two-week visit; two more weeks of waiting is exactly the delay we are trying to prevent. The second is to accept the parents' report that the stools look yellow. Pale stools are easily misread and can be intermittent, so look at the nappy yourself and use the card. The third is to be reassured by a falling total bilirubin. In the newborn the unconjugated fraction falls physiologically and can hide a rising direct fraction, so always ask for the direct value, and remember that a previously high direct bilirubin counts even if the next one looks better. The fourth is to declare an infant too old for a Kasai. One in five still keeps the liver; judge the liver, not the date, and discuss the case with a transplant centre. And one more: treating sixty days as a deadline that has already passed when the infant is sixty-two days old. The curve is continuous, and the next week matters as much as the last one did.

Return now to the seven-week-old girl with pale stools. The answer to the referring doctor is that next Wednesday is too late. She should be seen by hepatology this week, given vitamin K, and have her ultrasound and blood tests done in parallel, with a cholangiogram and Kasai within days if atresia remains likely, in a centre that does this operation often. A few other scenarios follow the same logic. A breastfed two-week-old who is mildly jaundiced needs a fractionated bilirubin that day, not at four weeks. A newborn whose screening direct bilirubin was above the reference range and is higher at two weeks should be treated as having atresia until proven otherwise and referred the same day. A ten-week-old with likely atresia, no ascites and a normal clotting profile after vitamin K

should have a Kasai now, with honest counselling that a transplant may be needed later. A sixteen-week-old with ascites and a coagulopathy that does not correct with vitamin K needs transplant assessment first. And if the local surgeon does one Kasai a year, transfer to a high-volume centre is worth a delay of days, but not of weeks.

So, tomorrow in clinic, carry these points. Survival with the native liver falls continuously with age at the Kasai, with no safe age and no futile age. In France it was 38 percent at twenty-five years after a first-month operation and 19 percent after three months; in Canada it was 49 percent at four years after an operation by thirty days and 23 percent after ninety days. Pooled data show that operating in the first month makes native-liver survival about one fifth more likely, and that later surgery roughly doubles the rate of reaching transplant. In the United States the median age is 57 days and almost half of infants are operated after sixty days. Stool cards in Taiwan moved more infants before sixty days and raised five-year survival from about 56 to about 89 percent, while blood screening in Texas found every case and brought surgery about twenty days earlier. Centre volume matters as much as age: three in five children kept their liver in larger British centres in the 1990s, against one in seven in small ones, and centralised care in England now achieves 55 percent clearance and 90 percent survival. Test the direct bilirubin at two to four weeks, treat one milligram per decilitre or more as urgent, run the work-up in parallel, and operate in an experienced centre, because in biliary atresia every week saved is liver saved.